Development of the gonads, male duct system

ALEXANDRIA NATIONAL UNIVERSITY
ANU MEDICAL SCHOOL PROGRAM

ILOs

- Describe the development of male & female gonads.
- Explain the role of hormones in early fetal life and its relevance to normal & abnormal sexual differentiation.
- Correlate the normal development of gonads to their congenital anomalies.
- Correlate the development of male duct system with their normal anatomy.

ALEXANDRIA NATIONAL UNIVERSITY
ANU MEDICAL SCHOOL PROGRAM

Development of the Genital system

Development of the gonads

Development of the genital ducts

Development of the external genitalia

ALEXANDRIA NATIONAL UNIVERSITY
ANU MEDICAL SCHOOL PROGRAM

Development of the gonads

The Indifferent Embryo

- The gender of the embryo is determined genetically at the time of fertilization.

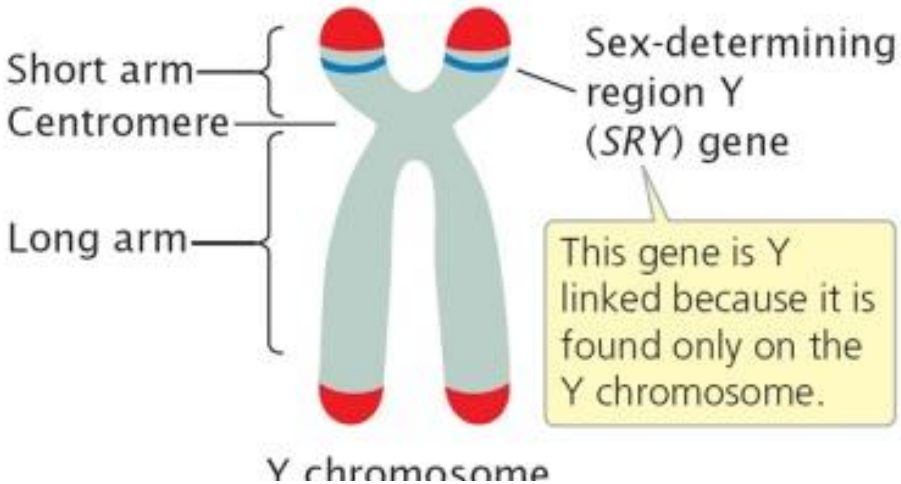
- However, the gonads do not acquire male or female morphological characteristics until the **seventh week** of development.





Development of the gonads

The Indifferent Embryo



- Gender determination is a complex process that involves many genes.
- The key to phenotypic sexual differentiation is the Y chromosome, which contains the **SRY (sex-determining region on Y) gene** which is testis-determining gene. This gene produces **SRY protein** (testis determining factor).
- If it is present → male.
- In its absence → female.

ANU MEDICAL SCHOOL PROGRAM